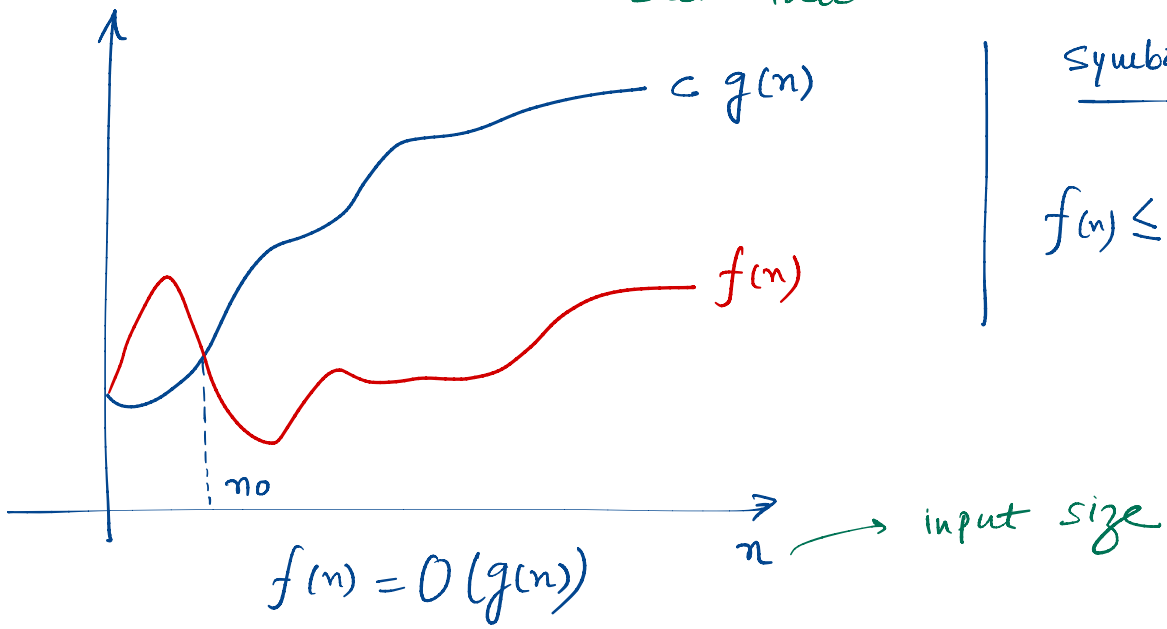
① $f(n) = O(g(n))$: Asymptotic upper bound

$$O(g(n)) = \{ f(n) : \exists \text{ constants } c > 0,\ n_0 > 0,\ \text{s.t. } 0 \le f(n) \le c\, g(n) \ \forall\, n > n_0 \}$$

There exists

for all

such that

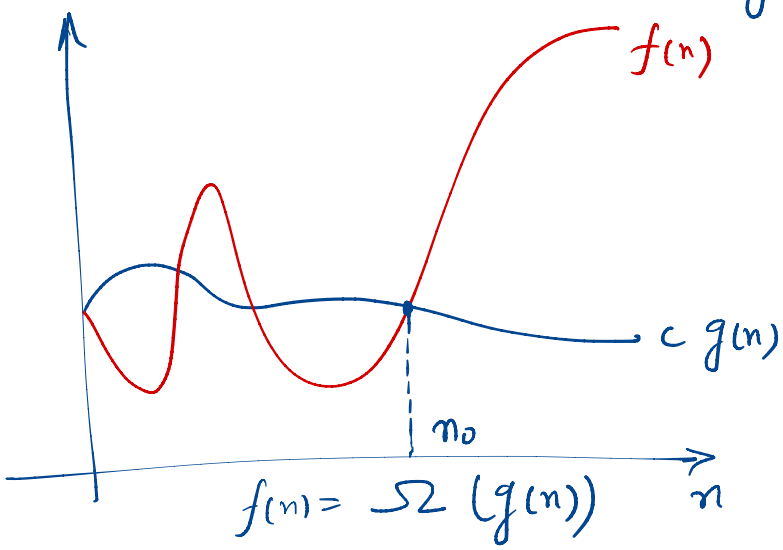


Symbolic

$f(n) \le g(n)$

② $f(n) = \Omega(g(n))$: Asymptotic lower bound

$$\Omega(g(n)) = \{ f(n) : \exists \text{ constants } c > 0,\ n_0 > 0 \text{ s.t. } 0 \le c\, g(n) \le f(n) \ \forall\, n > n_0 \}$$



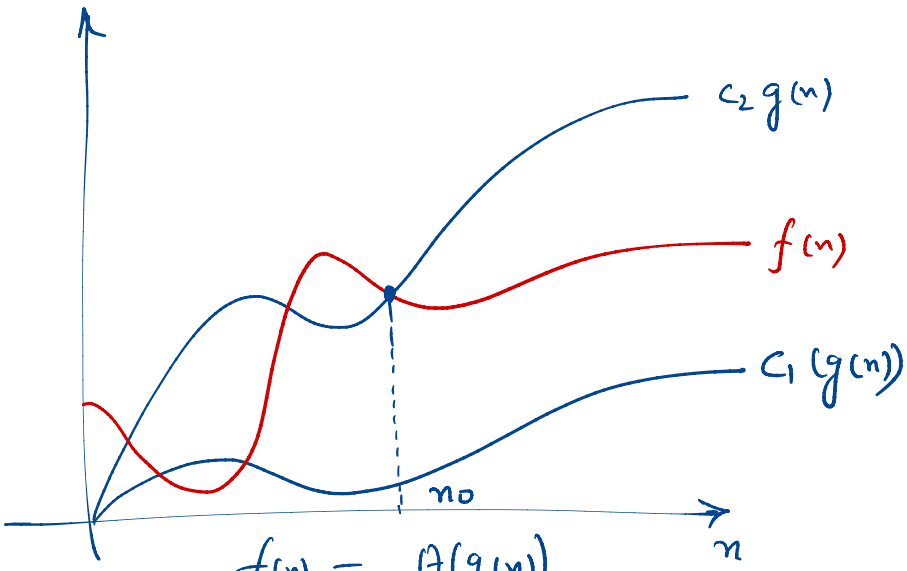
Symbolically

$f(n) \ge g(n)$

③ $f(n) = \Theta(g(n))$: Asymptotic tight bound

$$\Theta(g(n)) = O(g(n)) \cap \Omega(g(n))$$

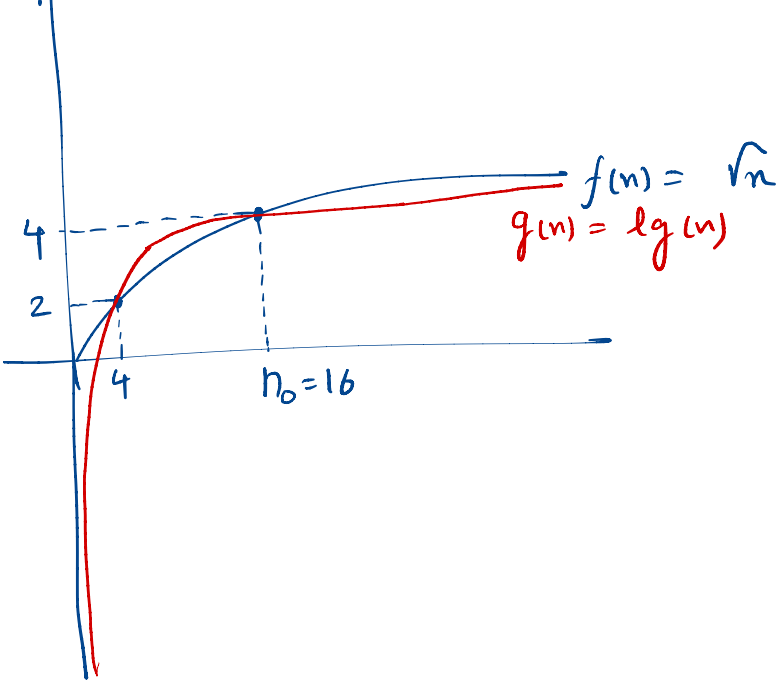
$$\Theta(g(n)) = \{ f(n) : \exists \text{ constants } c_1, c_2, n_0 > 0, \text{ s.t. } 0 \le c_1 g(n) \le f(n) \le c_2 g(n) \ \forall n > n_0 \}$$



$f(n) = \Theta(g(n))$

$\sqrt{n} = \Omega(\lg(n))$ $\quad c=1, \ n_0 = 16$

Ex. Asymptotic lower bound : Ω



Ex. Asymptotic tighter bound : Θ

$n\left(\frac{n}{2}-2\right)$

$f(n) = \frac{1}{2}n^2 - 2n$ $\qquad g(n) = n^2$ $\qquad c_2 = 3/4$

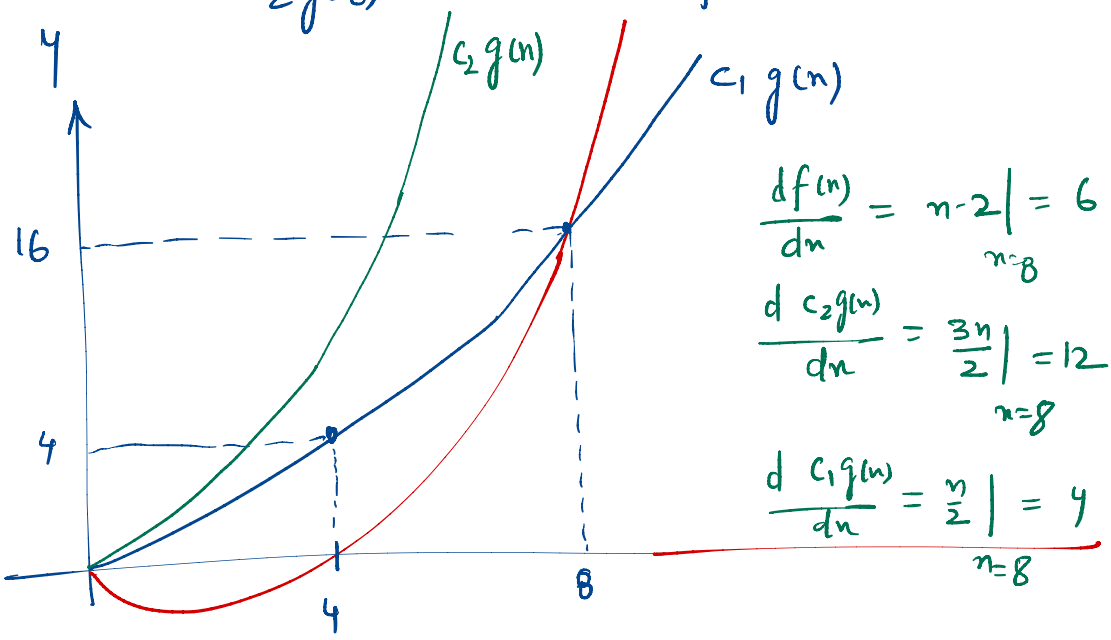
$c_1 = 1/4$

$c_1 g(n) = n^2/4$, $c_2 g(n) = \frac{3n^2}{4}$ $\qquad n_0 = 8$

$\downarrow \qquad \downarrow$

$c_1 g(n_0) = 16$ $\qquad c_2 g(n_0) = 48$ $\qquad f(n_0) = 32 - 16 = 16$

$c_1 g(4) = 4$



$\frac{df(n)}{dn} = n-2 \Big|_{n=8} = 6$

$\frac{d\, c_2 g(n)}{dn} = \frac{3n}{2} \Big|_{n=8} = 12$

$\frac{d\, c_1 g(n)}{dn} = \frac{n}{2} \Big|_{n=8} = 4$

Solving Recurrence Relations

3 methods:

1. Substitution method
2. Recursion tree method
3. master method